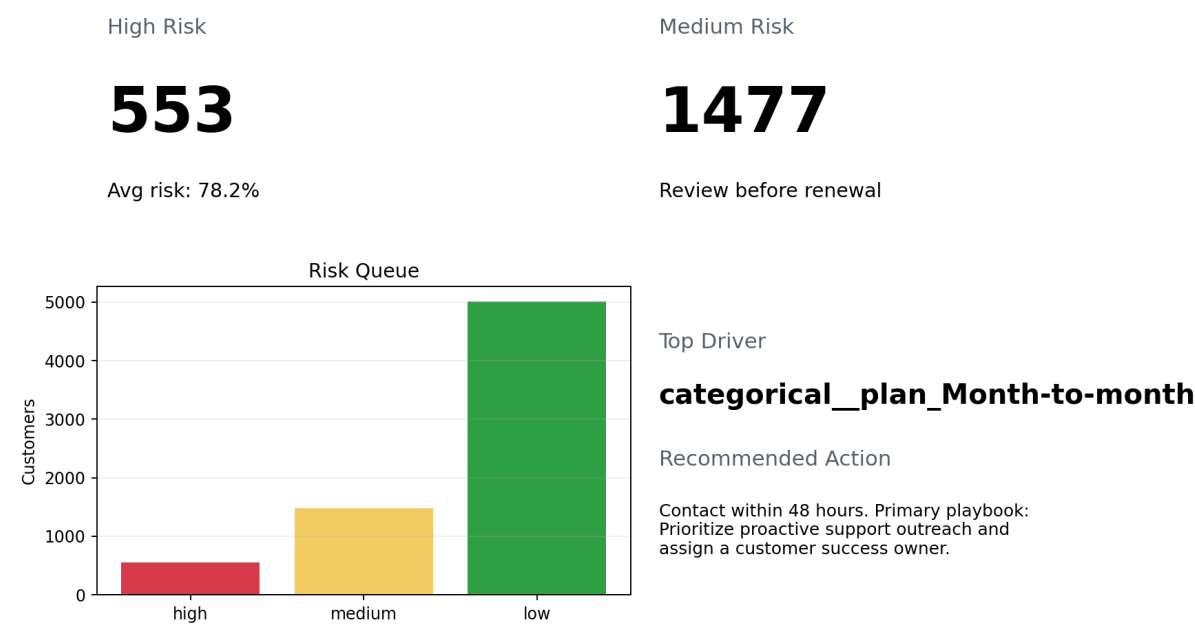


Churn Prediction + Retention Report

Executive summary: this report ranks customers by churn risk and turns model output into a prioritized retention action queue.

The model ranks customers by churn risk. The top-risk segment should be reviewed first because it concentrates the highest expected churn probability.

Churn Retention Executive Snapshot



Dataset Summary

Metric	Value
rows	7043
columns	11
observed_churn_rate	26.54%
duplicate_customer_ids	0
leakage_warnings	0

Model Metrics

Metric	Value
roc_auc	0.8435
pr_auc	0.6564
baseline_pr_auc	0.2655
accuracy	0.7908

Metric	Value
precision	0.6319
recall	0.508
f1	0.5632
confusion_matrix	[[1386, 166], [276, 285]]
brier_score	0.136
roc_auc_ci_low	0.8253
roc_auc_ci_high	0.8619
pr_auc_ci_low	0.6087
pr_auc_ci_high	0.6993

Risk Segment Breakdown

risk_segment	customers	avg_churn_probability
high	553	0.7818
medium	1477	0.5405
low	5013	0.1267

Lift Table Interpretation

The highest-risk 10% captures 28.9% of observed churners. The highest-risk 20% captures 48.1%. Review the 553 high-risk customers first.

decile	customers	churners	churn_rate	avg_churn_probability	cumulative_churners	cumulative_churn_rate
1	212	162	0.7642	0.7496	162	0.2888
2	211	108	0.5118	0.5772	270	0.4813
3	211	100	0.4739	0.4521	370	0.6595
4	211	77	0.3649	0.3273	447	0.7968
5	212	43	0.2028	0.2296	490	0.8734

Probability Evidence

Calibration and confidence intervals are computed from holdout customers only, so model evidence stays separate from the full customer action queue.

metric	estimate	ci_low	ci_high	bootstrap_samples
roc_auc	0.8435	0.8253	0.8619	300
pr_auc	0.6564	0.6087	0.6993	300

bin	customers	mean_predicted_probability	observed_churn_rate	absolute_gap
1	212	0.7496	0.7642	0.0145
2	211	0.5772	0.5118	0.0654
3	211	0.4521	0.4739	0.0219

bin	customers	mean_predicted_probability	observed_churn_rate	absolute_gap
4	211	0.3273	0.3649	0.0376
5	211	0.2299	0.2038	0.0261
6	212	0.1456	0.1462	0.0006

Top Recommended Actions

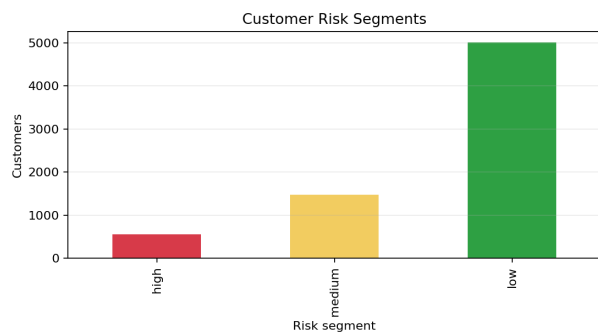
customer_id	risk_segment	churn_probability	likely_drivers	retention_priority
7216-EWTRS	high	0.9061	support_tickets, payments	1
6567-HOOPW	high	0.9058	support_tickets, payments	1
8580-AECUZ	high	0.9058	support_tickets, payments	1
9300-AGZNL	high	0.9054	support_tickets, payments	1
9497-QCMMS	high	0.9040	support_tickets, payments	1

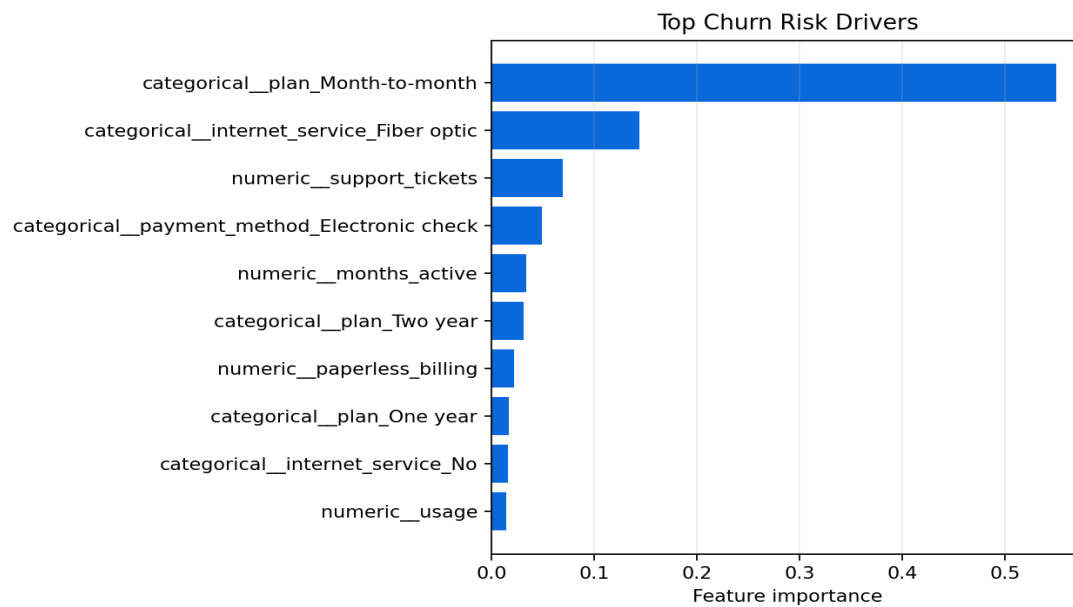
Detailed recommended_action text is delivered in retention_recommendations.csv so the PDF stays readable.

Recommended Retention Strategy

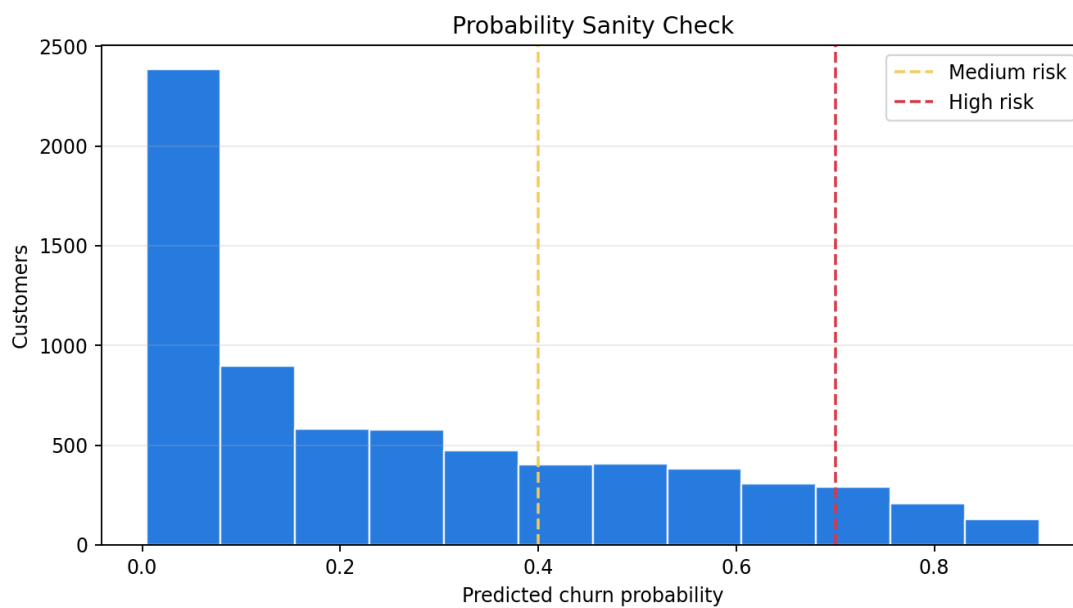
Use the retention recommendation CSV as the operating queue. High-risk customers should receive direct outreach; medium-risk customers should receive targeted education, usage review, or plan-fit checks.

Visual Evidence

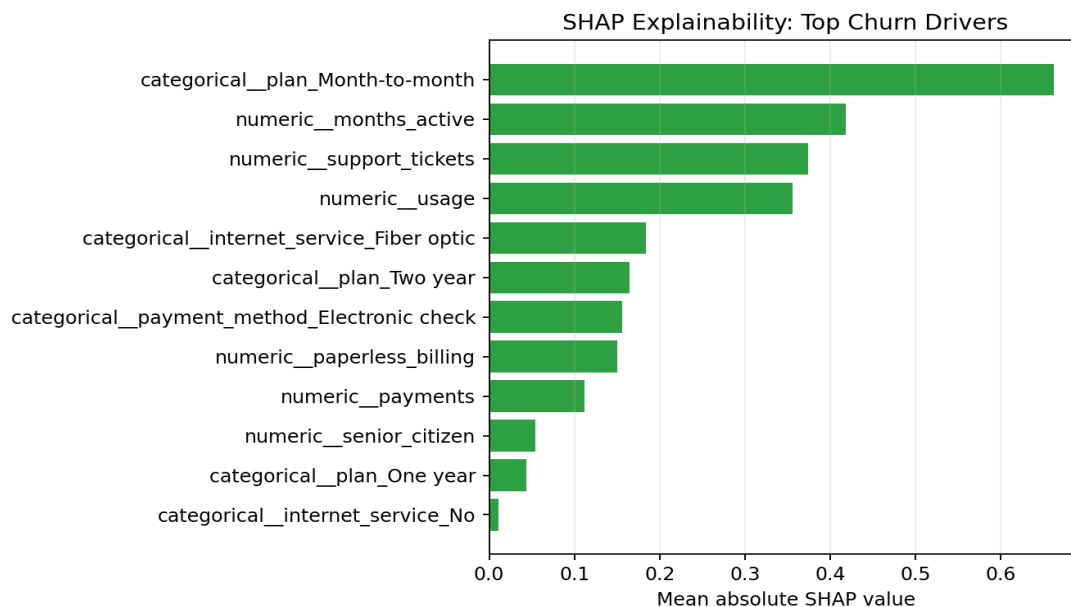




Probability Sanity Check



SHAP Explainability



Limitations

This report does not guarantee a specific accuracy number. Results depend on data volume, feature quality, label quality, and whether post-churn leakage columns were excluded before modeling.

Next Steps

Recommended next steps: review leakage warnings, confirm the business definition of churn, enrich weak features, run the action queue, and compare retention outcomes after the next billing cycle.

Scope and Privacy

This repository keeps raw private datasets out of version control. The checked-in sample artifacts are reproducible from the included public-data workflow and are meant to show the report shape without exposing private data.